Quiz 3 Reflection

SCIE 001 Physics

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Contents

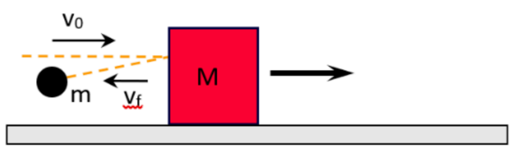
1. Collision Ball and Box 1

1.1. Specify Question

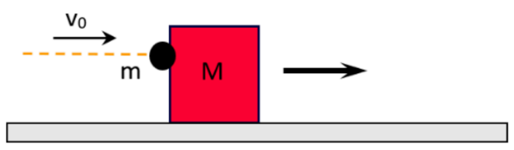
Question 5: Collision Ball and Box 1

Consider the following two cases:

Case 1



Case 2



In both cases, a ball of mass m is thrown horizontally with speed v_0 at a stationary box of mass M. The box slides without friction and all motion is horizontal.

In Case 1, the ball bounces off the box and after the collision the box is moving to the right with speed V_1 and the ball is moving to the left with speed v_f.

In Case 2, the ball sticks to the box and after the collision the box (with the ball stuck to it) is moving to the right with speed V_2.

Which of the following statements best describes V_1 and V_2?

- $V_2 > V_1$
- $V_2 = V_1$ ✕
- $V_1 > V_2$
- Depends on how much energy is lost during the collisions

✕ 0%

This question is complete and cannot be answered again.

1.2. Diagnosis Phase

I was going too fast. I misinterpreted the image and thought that in case 1 the velocity of the ball when it bounced back was in the same direction as the velocity of box. This made me think that if the momentums were equal, they must have the same velocity.

For case 1:

$$p_{\text{initial}} = p_{\text{final}}$$
$$mv_0 = mv_f + MV_1$$

For case 2:

$$p_{\text{initial}} = p_{\text{final}}$$
$$mv_0 = (m+M)V_2$$

Combining these:

$$mv_f + MV_1 = (m+M)V_2$$

I got stuck here - because I thought v_f was positive, I didn't have enough information. Since v_f was in the same direction as V_1, out of all 3 possibilities $(>, <, =)$, I assumed $v_f = V_1$ and factored.

$$mV_1 + MV_1 = (m+M)V_2$$
$$(m+M)V_1 = (m+M)V_2$$

So I said $V_1 = V_2$.

1.3. Correction Phase

In both elastic and inelastic collisions, momentum is conserved. So we can write out the equations for both sides (with proper signs) to show the relationship between V_1 and V_2 (explicitly showing negative vectors).

For the first case:

$$p_{\text{initial}} = p_{\text{final}}$$
$$mv_0 = m(-v_f) + MV_1$$

For the second case:

$$p_{\text{initial}} = p_{\text{final}}$$
$$mv_0 = (m+M)V_2$$

So:

$$(m+M)V_2 = m(-v_f) + MV_1$$

$$V_2 = \frac{-v_f m}{m+M} + \frac{M}{m+M}V_1$$

We know that V_2, V_1, m, and M are positive, and we have expressed V_2 as some negative number plus some fraction $0 < \frac{M}{m+M} < 1$ of V_1.

$$\frac{-v_f m}{m+M} + \frac{M}{m+M}V_1 < V_1$$
$$V_2 < V_1$$

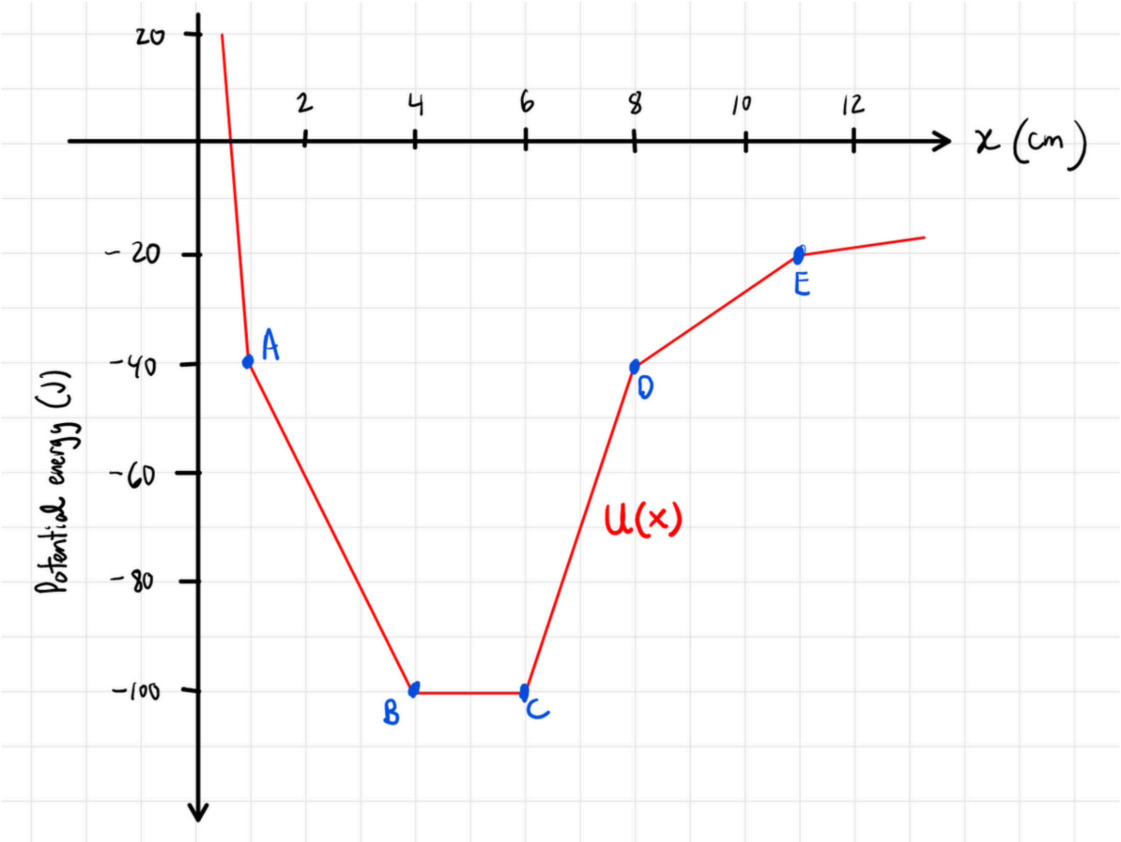
Therefore V_1 must be greater than V_2.

2. Interpreting an energy diagram

2.1. Specify Question

Question 9: Interpreting an energy diagram

A system is composed of two particles: the first is fixed at $x = 0$ cm and the second can move. The potential energy of the system $U(x)$ as a function of the particle separation x is shown below. At a certain point in time, the second particle is at point D and has kinetic energy $K = 10$ J.



a) What is the maximum kinetic energy of the second particle?

70 J

✓ 100%

b) At what separation is there a turning point?

5 cm

✗ 0%

c) Over which segment does the force applied on the second particle have the largest magnitude:

CD

✓ 100%

d) What type of interaction best characterizes this system?

attractive or repulsive depending on x

✓ 100%

This question is complete and cannot be answered again.

2.2. Diagnosis Phase

The issue was that I simply did not know what a turning point was. It seems I had zoned out in the lecture covering it, and not reviewed it before the test.

I incorrectly assumed a turning point was the point at which the force changed direction. Using the fact that force is equal to derivative of potential energy with respect to distance, the slope of the graph indicates that the second particle is attracted to the first from 13 cm to about 5 cm where it is then repulsed up to 0 cm.

So I answered 5 cm.

2.3. Correction Phase

A turning point is when the kinetic energy (and hence velocity) of an object is zero. In this system the turning point represents a separation of the particles where the moving particle stops and reverses direction as it oscillates about the equilibrium point.

Expressing potential and kinetic energy as functions of distance and keeping total mechanical energy constant:

$$\begin{aligned}
E_{\text{mechanical}} &= \text{constant} \\
&= U(x) + K(x) \\
&= U(D_x) + K(D_x) \qquad D_x = 8\text{ cm} \\
&= U(8\text{ cm}) + K(8\text{ cm}) \\
E_{\text{mechanical}} &= -40J + 10J = -30J
\end{aligned}$$

To find where $K(x) = 0$:

$$\begin{aligned}
E_{\text{mechanical}} &= U(x) + K(x) \qquad K(x) = 0 \\
-30J &= U(x)
\end{aligned}$$

Hence, there is a turning point where $U(x) = -30$.

Therefore there is a turning point between 9 cm and 10 cm. (Note: 0.9 cm was not an option)